

Natural compounds in the human diet and their ability to bind mutagens prevents DNA-mutagen intercalation.

Human diet may contain many mutagenic or carcinogenic aromatic compounds as well as some beneficial physiologically active dietary components, especially plant food phytochemicals, which act as mutagenesis or carcinogenesis inhibitors. This study compared the binding properties of natural compounds in the human diet (caffeine, theophylline, theobromine, and resveratrol) with a water-soluble derivative of chlorophyll to bind to acridine orange, a known mutagen. An analysis was conducted to determine which substances were effective binding agents and may thus be useful in prevention of chemical-induced mutagenesis and carcinogenesis. Data indicated that in order to bind 50% of the mutagen in a complex, less than twice the concentration of chlorophyllin was needed, the resveratrol concentration was 20-fold higher, while a 1000-fold or even 10,000-fold excess of xanthines were required to bind acridine orange.